

Aprille J. Xi

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Education

Carnegie Mellon University

Bachelor of Computer Science, Machine Learning Concentration

Pittsburgh, PA

2026

Relevant Courses

Algorithms, Distributed Systems, Deep Learning, Computer Systems, Functional Programming, Modern Regression, Machine Learning, Computer Security, Program Analysis, ML w/ Large Datasets, Generative AI

Experience

Tools for Humanity – Applied Cryptographic Research

San Francisco, CA

Engineering Fellow

June 2026 – Present

- Designed and built a Rust service on AWS Nitro Enclaves to batch, encrypt, sign, attest, and verify correlated randomness for secure multiparty computation (MPC) between host instance/enclave [\[GitHub\]](#)
- Implemented a configurable k -of- n replicated secret-sharing (RSS) MPC framework in Rust, including PRNG-optimized multiplication and MSB extraction via optimized binary adder circuits [\[GitHub\]](#)
- Researched trusted execution environments (TEEs) security guarantees/risks and led technical reading groups and presentations on TEEs, AWS Nitro System + Nitro Enclaves security, and k -of- n RSS

SNAP Lab – Cryptosystems Research

Pittsburgh, PA

Undergraduate Researcher

December 2024 – May 2026

- Explored bootstrap placement for fully homomorphic encryption (FHE, CKKS) for large neural networks under Dr. Wenting Zheng at CMU CSD's Cryptosystems Group in SNAP Lab
- Built an optimized, fully correct, memory-safe parser and executor using the FHE Lattigo Library in Go
- Helped build end-to-end pipeline and conducted numerous cost-modeling tests in Python and Go
- Reduced average and peak memory usage by 20%, and runtime by 9%

Learnlab – Machine Learning Research

Pittsburgh, PA

Research Assistant

August 2024 – May 2026

- Built Chrome extension for LLM screen analysis under Prof. Ken Koedinger's LearnLab at CMU HCI
- Extended the Chrome extension into LearnLab's VTutor platform with vision-based screen analysis and UI features in TypeScript and JavaScript. Work showcased in demos and accepted in various papers.

Projects

MARAGE: Multi-Agent RAG Hallucination Mitigation — [GitHub](#)

August 2026

- Conceived, researched, designed, and implemented a multi-agent retrieval-augmented generation (RAG) system for hallucination mitigation, integrating FAISS retrieval and evidence-grounded response revision
- Benchmarked the system on generated hallucination-inducing prompts, improving the Agent 1→2 correction metric from 10.85% to 13.11% and reducing hand-labeled hallucination rate from 46% to 34%

Content Curator for NSF AI-SDM

December 2023 – May 2026

- Wrote interviews for AI professionals on ethics, public understanding, and societal impacts

Empathy (TartanHacks) — [GitHub](#)

February 2023

- Project Leader, Programmed Flask API w/ trained CNN model for facial emotional recognition

Papers

E. Chen, X. Tang, **Aprille Xi**, C. Lin, *et al.*,

Vtutor for high-impact tutoring at scale: Managing engagement and real-time multi-screen monitoring with p2p connections, *Preprint*, 2025. [Online]. Available: <https://dl.acm.org/doi/pdf/10.1145/3698205.3733948>

Skills

- Languages:** Rust, Go, Python, C, Java, SML
- Tools:** Linux, Git, AWS, Cargo, Docker

Awards

- TartanHacks:** Top 5 Overall, Best AI Project (2023), **HackCMU:** 3rd Place Overall (2022)